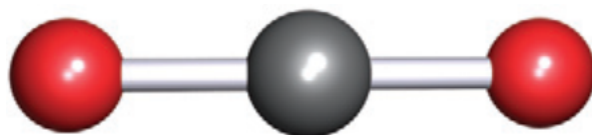


Lecture (11)

Carbon dioxide

Carbon dioxide is a colorless, odorless gas with a faint acid taste. Under normal circumstances, the gas is nontoxic, although at high concentrations it interferes with respiration. Carbon dioxide does not support most combustions, which makes CO_2 useful as a fire extinguisher.

Carbon in CO_2 is sp hybridized. The two sp orbitals form two bonds with two oxygen atoms. The two p orbitals not included in hybridization give rise to two p bonds. Carbon dioxide ($\text{O} = \text{C} = \text{O}$) is a linear molecule with carbon-oxygen bond equal to 115 pm, which is intermediate between those calculated for carbon-oxygen double and triple bond. Ideal angles equal 180° . It is thus considered to be the resonance hybrid of the following structures:



Liquid and solid carbon dioxide are used as refrigerants, and the gas is used to make carbonated beverages.

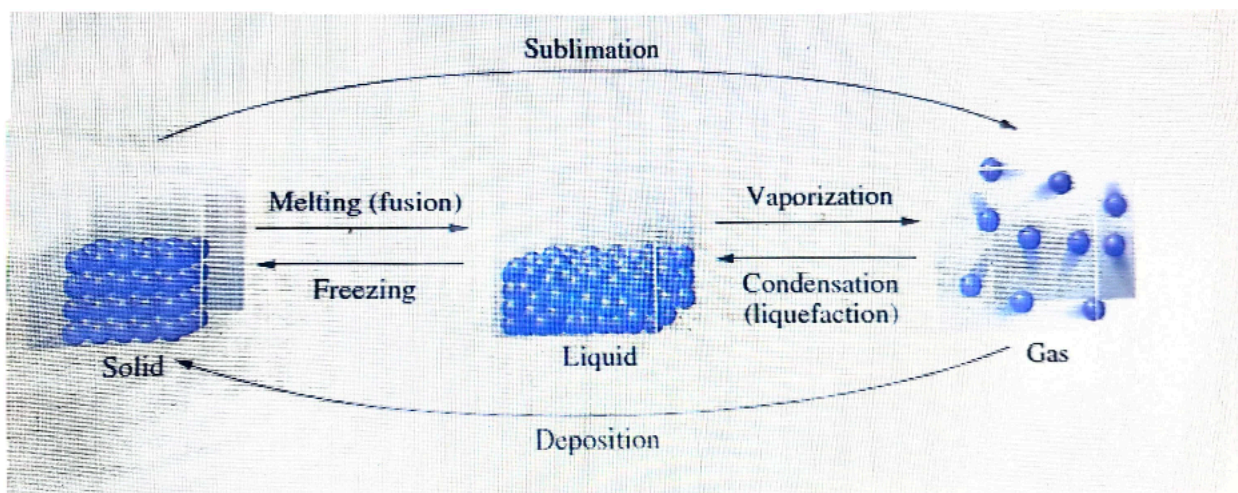
A carbonated beverage is produced by dissolving carbon dioxide in beverage solution under pressure. More carbon dioxide dissolves at the higher pressure than otherwise. When this pressure is suddenly released, carbon dioxide is less soluble, and the excess bubbles from the solution.



Dry ice is solid carbon dioxide, CO_2 , whose temperature is about -78°C . It passes directly from the solid to the gaseous state. The white plumes are composed of ice crystals (H_2O , not CO_2) formed by condensation from the cold air. Solid carbon dioxide is known as dry ice and is used as a refrigerant.



carbon dioxide, changes to the liquid state, the phase transition is often

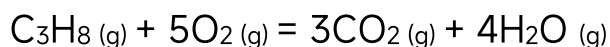
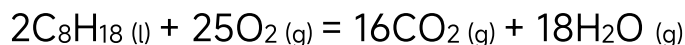


referred to as *liquefaction*. The following diagram summarizes these phase transitions

Preparation of Carbon dioxide:

- Carbon burns in an excess of oxygen to give carbon dioxide, which

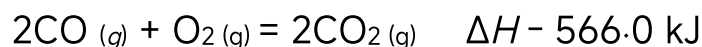
is the acid anhydride of carbonic acid (that is, carbon dioxide produces carbonic acid when it reacts with water). A hydrocarbon such as octane, C_8H_{18} , burns to give carbon dioxide and water.



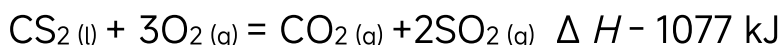
- Calcium carbonate at high temperatures decomposes into calcium oxide and carbon dioxide.



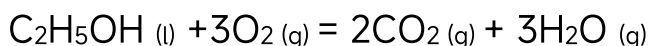
- react carbon monoxide with oxygen:



- react carbon sulfide with oxygen:



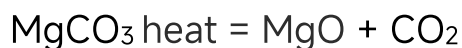
- oxidation of ethanol :



- Carbonates decompose when heated. For example,

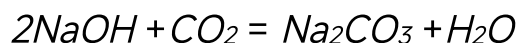
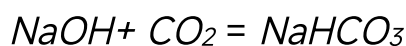


- **Carbon dioxide** can be prepared by any of the following reactions.



Reactions of Carbon dioxide:

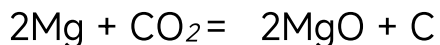
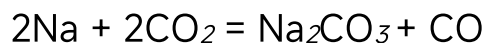
- *Sodium hydroxide reacts with carbon dioxide according to the following equations:*



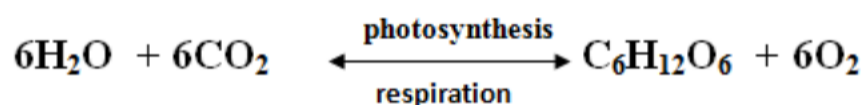
- Carbon dioxide is an acidic oxide.



- With reactive metals, it is reduced to CO.



- Carbon dioxide is absorbed by green plants in the presence of sunlight and is ultimately transformed into starch and cellulose in the chloroplast. This process is known as photosynthesis.



While respiration is the opposite process, where oxygen react with glucose to form carbon dioxide and water and energy.

Helium

Helium (${}^4_2\text{He} = 1s^2$) is Colourless noble gas with boiling 4.2 c. During a solar eclipse in 1868, a new emission line, matching no known element, was found in the spectrum of the solar corona. J. N. Locklear and E. Frankland proposed the existence of a new element named, appropriately, helium (Greek, helios, sun). Helium is fairly rare on Earth, but it is the second most abundant element in the universe (76% H, 23% He) and is a major component of stars.

Commercially, helium is obtained from natural gas. Helium is used as an inert atmosphere for arc welding, in weather and also used to fill balloons and air ships because of its non-inflammable nature.

Helium in gas mixtures (An O_2/He mixture is used in place of O_2/N_2) used in deep-sea diving, where it gives voices a higher pitch, but is less

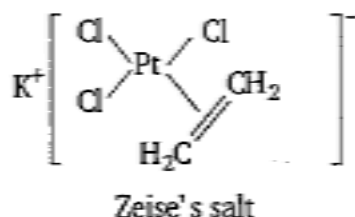
soluble in blood than nitrogen. Recently, liquid helium (with a boiling point of 4.2 K) has increasingly been used as a coolant for superconducting magnets in NMR instruments, including those used for medical imaging.

Helium is also used as a heat transfer agent in gas-cooled nuclear reactors, for which it has the advantages of being non-corrosive and of not becoming radioactive under irradiation.

Organometallic Chemistry

Organometallic chemistry is the study of chemical compounds (organometallic compounds) containing one or more bonds between carbon and a metal.

The first organometallic compound of the transition metals was Zeise's salt, $K[Pt(C_2H_4)Cl_3]$.



Classification of organometallic compounds:

The organometallic compounds are classified in two type depend on the Structure as follows :

- **simple organometallic compounds :**

structure	IUPAC name
$(CH_3)_4Pb$	Tetramethyllead
$(CH_3)_3SnH$	Trimethylstannane

- **Mixed organometallic compounds :**

structure	IUPAC name
CH_3MgBr	Methyl magnesium bromide
$(CH_3)_2SnCl_2$	Dichloro(dimethyl)stannane

The organometallic compounds are classified in two types depend

on the place of the metal in periodic table as follows :

- **Main group organometallic compounds:**

CH_3Li (Methyl lithium)

CH_3MgBr (methyl magnesium bromide)

$(\text{CH}_3)_4\text{Pb}$ (Tetramethyllead).

- **d-block and f-block elements:**

$\text{Fe}(\eta^5\text{-C}_5\text{H}_5)_2$ Ferrocene (η^5 Pentahaptoligand)

$\text{U}(\eta^8\text{-C}_8\text{H}_8)_2$ Uranocene (η^8 Octahaptoligand)

Nomenclature of organometallic compounds:

Nomenclature depend of the type of metal and a part of alkyl group. Name of organometallic = name of alkyl group + metal name.

$(\text{CH}_3)_4\text{Si}$	Tetramethylsilane Tetramethylsilicon
$\text{B}(\text{CH}_2\text{CH}_3)_3$	Triethylborane
$\text{CH}_3\text{CH}_2\text{MgBr}$	Ethyl magnesium bromide
$(\text{CH}_3)_3\text{Al}$	Trimethyl aluminum
$(\text{CH}_3)_3\text{SiCH}_2\text{CH}_3$	Ethyl trimethyl silicon
$\text{CH}_3\text{CH}_2\text{CH}_2\text{MgBr}$	Propyl magnesium bromide

Organometallic Compounds as Drugs : The first organometallic compound found for anticancer activity was Cp_2TiCl_2 [Fig. 1.1(a)]. Another important organometallic compound is ferroquine (FQ) [Fig. 1.1(b)] which is used as antimalarial drug.

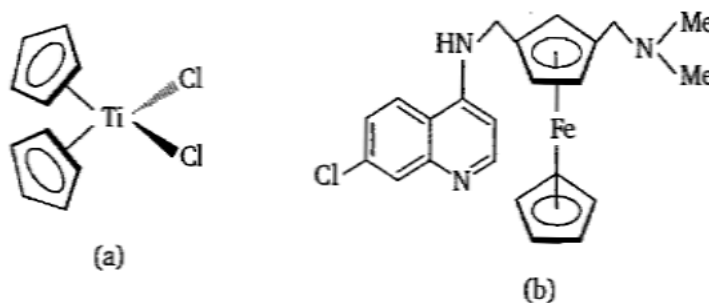


Fig. 1.1 Structure of (a) Cp_2TiCl_2 (b) Ferroquine

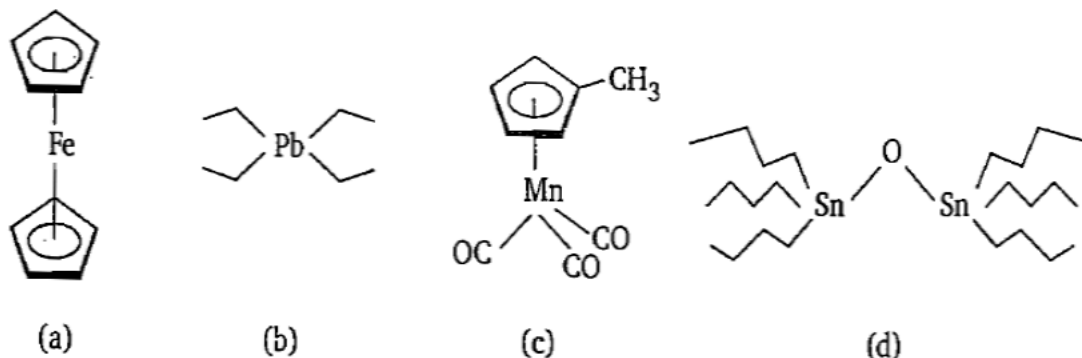
Application of organometallic compound:

- (1) organometallic compounds as reagents: some organometallic compounds especially the alkali and alkaline earth metal compounds such as $n\text{-Bu-Li}$, $s\text{-Bu-Li}$, $t\text{-BuLi}$, Grignard reagent (RMgX) etc are

used as reagent.

(2) organometallic compounds as catalysts: the organometallic compounds are being routinely used as catalysis for the synthesis of organic compound.

(3) Organometallic compounds as additives : Organometallic compounds have find applications as bulk additives for modifying the existing properties of compounds. Ferrocene is readily soluble in liquid fuels, air stable, non-toxic and usually thermally stable. Ferrocene is, therefore, added to liquid fuel to reduce carbonaceous particulate matter like soot emitted by diesel engines. The carbonaceous particulates are health hazard. Tetraethyllead [Fig. 1.2(b)] was used as antiknocking agent in the gasoline. TEL helps the gasoline to burn slowly and smoothly and prevents knocking. Combustion of TEL along with gasoline forms particles of Pb and PbO which are very harmful for health. Now, therefore methylcyclopentadienylmanganesetricarbonyl (MMT) [Fig. 1.2(c)] is used in place of TEL as anti-knocking agent.



Hapticity

A single organic ligand may interact with a central metal atom using one or more of its atoms simultaneously. The number of atoms in a ligand attached to the metal atom is denoted by the prefix η (the Greek letter eta) followed by a superscript indicating the number of ligand atoms attached to the metal atom. This is called hapticity. Most ligands attach through one atom only, therefore, they are called as monohapto (η^1). Cyclopentadienyl ligand, C_5H_5 or Cp, for example, can attach to metal atom through one, three or five carbon atoms. Therefore, it may act as mono (η^1), tri (η^3) or pentahapto (η^5)-ligand (Fig. 1.3).

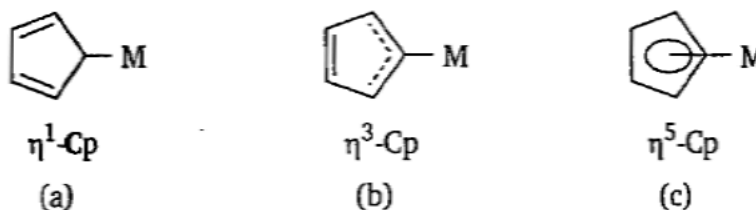
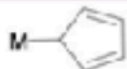
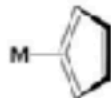


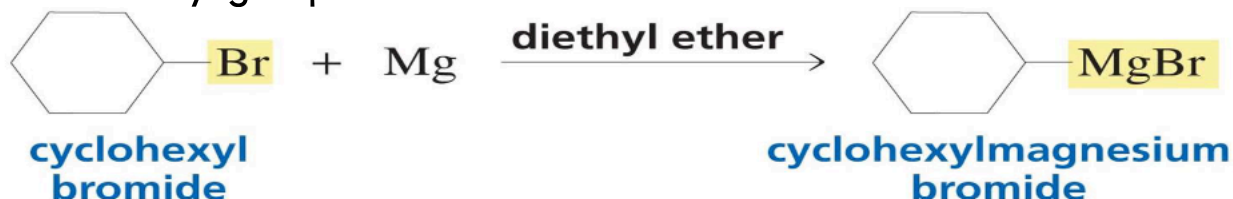
Fig. 1.3

Number of Bonding Position $x \rightarrow \eta^x$

Number of Bonding Positions	Formula	Name	
1	$\eta^1\text{-C}_5\text{H}_5$	Monohaptocyclopentadienyl	
3	$\eta^3\text{-C}_5\text{H}_5$	Trihaptocyclopentadienyl	
5	$\eta^5\text{-C}_5\text{H}_5$	pentahaptocyclopentadienyl	

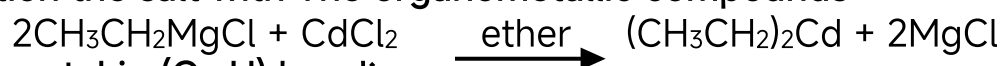
Preparing of organometallic compounds:

1- from alkyl group with the metal:



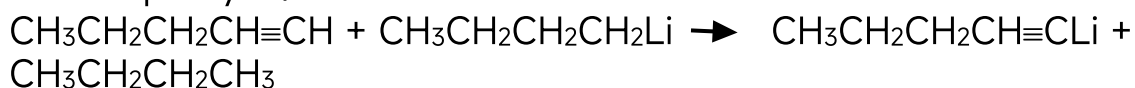
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2- reaction the salt with The organometallic compounds:



3- inter metal in (C-H) bond:

reaction of alkyne with lithium alkyl to produce lithium alkenyl (1-lithio-1-pentyne) and alkane



4- special method such as addition of alkene to boron hydride:

